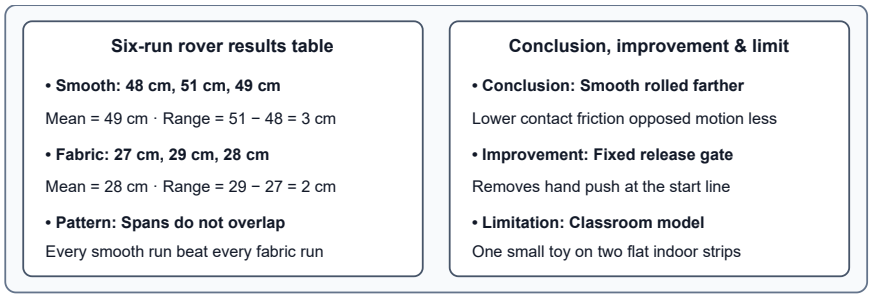
Reference sheets

Reference A · Six-run rover results table, means, ranges, and friction

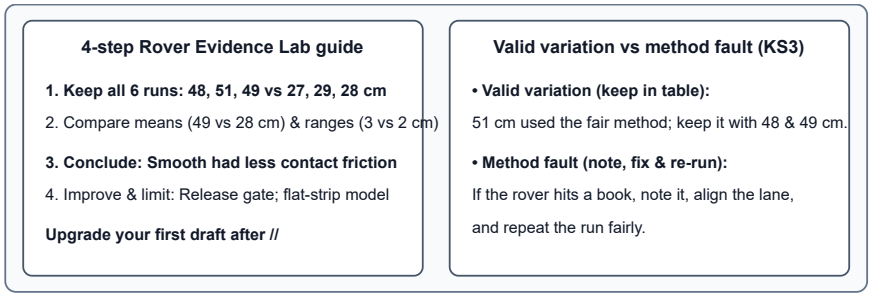


Six-run rover results table. It shows smooth and fabric runs, means, ranges, and friction.

Reference only. Do not write on this sheet. Use all six values to support your conclusion.

Surface condition.	Run 1, Run 2, Run 3 (cm).	Mean and range (cm).	Key Stage 3 science reason and limit.
1. Smooth surface.	48 cm, 51 cm, 49 cm (keep all three).	Mean = 49 cm. Range = 51 − 48 = 3 cm.	Lower contact friction opposed rover motion less on smooth surface.
2. Fabric surface.	27 cm, 29 cm, 28 cm (keep all three).	Mean = 28 cm. Range = 29 − 27 = 2 cm.	Greater contact friction opposed rover motion more on fabric.
3. Pattern across conditions.	Smooth span (48–51 cm) vs Fabric span (27–29 cm).	Difference between means = 21 cm.	Spans do not overlap. Limit: tested one small rover on two flat strips.

Reference B · Four-step Rover Evidence Lab frame and // upgrade



Four-step Rover Evidence Lab guide. It shows keeping six runs, comparing means, concluding, and upgrading with //.

Lab step.	Sentence stem for your book.	What to check before and after //.
1. Evidence (Keep all 6 runs).	Smooth runs were ___ cm (mean ___ cm, range ___ cm). Fabric runs were ___ cm (mean ___ cm, range ___ cm).	Did you keep all three repeats for each surface with cm units?

Lab step.	Sentence stem for your book.	What to check before and after //.
2. Conclusion (Link friction).	In this test, the rover rolled farther on ___ because ___ created less contact friction.	Did you name contact friction and bound the claim to this test?
3. Improvement & limitation.	To improve the method, I would add a ___. One model limitation is ___.	Did you distinguish a method improvement from a model limitation?
4. Improve with //.	Keep your first draft visible. Add // and your upgrade in the same colour.	Did you respond to the check question without erasing your first idea?

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival drawing. State the independent and dependent variables in the six-run rover test.

Help. Check Box 1 for what changed and what was measured.

2. Examine Box 1 and Box 2. Why do we keep the 51 cm smooth run instead of deleting it?

Help. Think about keeping all valid repeats.

3. Examine Box 2. Compare the mean distance on smooth surface (49 cm) with fabric (28 cm) using friction.

Help. Link the two mean distances to contact friction.

4. Examine Box 3. State one method improvement and one model limitation.

Help. Check the two bullets in Box 3.

You do tasks

1. Use Reference A. Compare all six runs (Smooth: 48, 51, 49 cm; Fabric: 27, 29, 28 cm). State the mean and range for both surfaces, and explain why 51 cm is kept.

Staff I ☐ S ☐ C ☐

Help. Include the mean and range for each condition.

2. Use Reference A and Reference B. Write an evidence conclusion explaining why smooth runs went farther than fabric runs. **Staff I ☐ S ☐ C ☐**

Help. Link the centimetre evidence to contact friction.

3. Use Reference B. State one method improvement and one model limitation. Then add one // upgrade after an adult check. **Staff I ☐ S ☐ C ☐**

Help. Check Steps 3 and 4 on Reference B.

Optional participation-loop prompt. Share your conclusion and improvement with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Calculate the range for smooth runs (48, 51, 49 cm) and fabric runs (27, 29, 28 cm).

2. State your conclusion for the six-run rover table using means and contact friction.

3. State one method improvement and one model limitation for this rover test.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt